linked with inflation. Perhaps it is no surprise, then, that most commodities turn out not to have any relationship with inflation.

In the real world, of course, gold and oil prices do not follow random walks. The perfect-world case is still useful because it allows us to see the effect of production frictions and how the behavior of investors and producers causes commodity prices to change in a predictable fashion. Today's economic models show that commodity prices reflect production costs. These vary over time, and, once you start production, you can't get your money back (they are *irreversible investments*), which impart an option value to delay production in the presence of uncertainty. Commodity prices also are driven by supply shocks to current and future substitutes, technological change—for example, a clean energy breakthrough that makes oil obsolete—and time-varying storage costs.[19] Some papers emphasize the role of speculators and other types of investors.[20] Speculators, for example, can cause commodity prices to temporarily swing away from fundamentals. Many of these factors do not have a direct bearing on inflation.

That many factors influence commodity prices means plenty of scope for active management in commodity investments. Commodity markets exhibit the same value and momentum effects, and other cross-sectional predictability phenomena, as equities and bonds.[21] Commodity Trading Advisors (CTAs), in fact, are often a byword for momentum-style trading. These value and momentum effects are manifestations of standard investment factors (see Chapter 7). In addition, commodity prices respond to hedging pressure, relative scarcity, and demand and supply imbalances, as predicted by economic models.[22] Open interest and other asset prices, including exchange rates, also forecast commodity prices.[23]

5.2. COMMODITY FACTORS

Commodities are usually not held physically by financial investors because of storage costs. Instead, investors gain commodity exposure through futures markets. An exception is the small group of investors who directly hold precious metals, but even in this case most investors seeking precious metal exposure (directly or indirectly through intermediated funds) invest through futures. Even for producers, the storage costs of some commodities, like electricity, are close to infinity.

Table 11.6 lists the performance of various commodity futures investments from the Goldman Sachs Commodity Index (GSCI) along with their correlations

[19] Carlson, Khokher, and Titman (2007) show that these effects give rise to stochastic volatility and determine the shape of the forward curves.

[20] Keynes (1923) and Hicks (1939) developed the first model of this type. Another seminal paper is Deaton and Laroque (1992).

[21] See Asness, Moskowitz, and Pedersen (2013).

[22] An important paper in this literature is Bessembinder (1992). Relative scarcity or demand in one commodity also spills over into other commodities, as Casassus, Liu, and Tang (2013) show.

[23] See Chen, Rogoff and Rossi (2010) and Hong and Yogo (2012).